

Faculty of Science Course Syllabus

Department of Mathematics and Statistics

MATH 2001 - Intermediate Calculus I

SUMMER 2026

Dalhousie University operates in the unceded territories of the Mi'kmaw, Wolastoqey, and Peskotomuhkati Peoples. These sovereign nations hold inherent rights as the original peoples of these lands, and we each carry collective obligations under the Peace and Friendship Treaties. Section 35 of the Constitution Act, 1982, recognizes and affirms Aboriginal and Treaty rights in Canada. We recognize that African Nova Scotians are a distinct people whose histories, legacies, and contributions have enriched the part of Mi'kma'ki known as Nova Scotia for over 400 years.

Course Code: MATH 2001
 Team: Daniel Teixeira (instructor)
 Jonathan Tot (TA)
 Iresha (marker)
 Brightspace: <https://dal.brightspace.com/d2l/home/435902>
 Lectures: <https://tinyurl.com/math2001teams>
 Day & Time: MWF @ 3:05–4:55 pm (Halifax time)
 (you can come to lectures or watch the recordings)
 dates: May 11–June 26, 2026
 email: daniel.teixeira@dal.ca
get help!
drop-in: after class until 6 pm (Halifax time)
appointments: <https://tinyurl.com/math2001booking>

Prerequisite. MATH 1010.

Course Description. Topics include review of parametric equations, polar coordinates, conic sections, coordinate systems and vectors, dot product and cross product, vector functions, derivatives and integrals of vector functions, arc length and curvature, functions of several variables and partial derivatives, directional derivatives and double and triple integrals.

Workbook. Fill-in-the-blank notes, by Deni Salja. Available at <https://tinyurl.com/math2001workbook> and on Brightspace.

Textbook. Calculus of Several Variables, by Serge Lang.

Notes. MATH 2001 notes, by Marcello Lanfranchi.

(TENTATIVE) SCHEDULE

We will fill our Multivariable Calculus worksheet in our lectures. The numbering below refers to those notes.

- week 1: vectors, parametric curves (chapters 0 & 1)
- week 2: multivariable functions, partial derivatives, gradient (chapter 2)
- week 3: chain rule, potential functions, path integrals (chapters 3 & 4)
- week 4: Taylor series, maximum and minimum (chapters 5 & 6)
- week 5: Lagrange multipliers, total derivative (chapters 6 & 7)
- week 6: double and triple integrals (chapter 8)
- week 7: (buffer/review) $\rightarrow$ final presentation this week

COURSE CONTENT

This is the third introductory course at Dalhousie about *Calculus*, which is the mathematical study of *continuous change*.

In your previous Calculus courses (MATH1000 OR MATH1215, or MATH1010) you studied **continuous functions one variable**, i.e. one input and one output, i.e. functions of the form $f: \mathbb{R} \rightarrow \mathbb{R}$ (the symbol $\mathbb{R}$ denotes the set of real numbers). In trying to understand such functions, we decided that **derivatives** were a good concept to describe rates of change, and **integrals** worked well in describing the area under a function.

In this course, we will study **continuous functions in many variables**. For instance, the function which inputs “coordinates in a map of Halifax” and outputs “altitude” can be understood as a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ (two-inputs-one-output). On the other hand, the function describing that inputs “time” and outputs “the coordinates of my bus on Google Maps” is a function $\gamma: \mathbb{R} \rightarrow \mathbb{R}^2$ (one-input-one-output). Like in your other Calculus courses, the goal is to “understand” functions like these, in a broad sense of the word.

We will use the language of **vectors** to describe lists of n variables, i.e. points in $\mathbb{R}^n$. We start by studying **parametric curves** ($\mathbb{R} \rightarrow \mathbb{R}^n$) and **multivariable functions** ($\mathbb{R}^n \rightarrow \mathbb{R}$). In trying to describe rates of change for multivariable functions, we develop the concepts of **partial derivatives** and **gradient**.

Given a parametric curve $\gamma: \mathbb{R} \rightarrow \mathbb{R}^n$ and a multivariable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, we study a **chain rule** for their composite $f \circ \gamma$. This chain rule motivates **potential functions** as “antiderivatives” for vector fields, and a concept of **path integral** for which we prove a version of the Fundamental Theorem of Calculus. We also provide a notion of **Taylor series** to multivariable function, i.e. approximations in terms of derivatives.

We find that the gradient is related to the **maximum and minimum** of a multivariable function. In fact, we learn the algorithm of **Lagrange multipliers** to find those max/min.

We define the **total derivative** of functions $\mathbb{R}^n \rightarrow \mathbb{R}^m$ (which is actually a matrix!), and end by calculating **double and triple integrals** (of 2d and 3d multivariable functions, respectively).

(Going on into your fourth and final Calculus course, you would study beautiful Fundamental Theorems of Calculus for functions $\mathbb{R}^n \rightarrow \mathbb{R}^m$.)

GRADING SYSTEM

Your grade is determined by the quantity of your assignments that are judged to be of a SATISFACTORY level of quality, and by your participation.

There is a two-step process for determining your final grade:

- (1) Your assignments determine your "letter grade", which is A, B, C, D, or F.
 - (2) Your participation points, obtained by engaging with the teaching team or with your peers. They determine if your letter grade should have a "plus" or a "minus" on it.¹
- (W) You must pass a final presentation to pass the class.

Details are below. Keep track of your progress, but you can always confirm it by asking your instructor.

Step 1: Letter Grade.

	to earn:	you must do:
A		8 WeBWork assignments (90%+)
		5 problem sets
		3 projects
		4 check-ins
		final presentation
B		6 WeBWork assignments (90%+)
		3 problem sets
		1 project
		2 check-ins
C		final presentation
		6 WeBWork assignments (90%+)
		3 problem sets
D		final presentation
		4 WeBWork assignments (90%+)
F		final presentation
		nothing

Step 2: Plus or Minus.

	to earn:	you must:
+		earn at least 20 participation points
-		earn less than 5 participation points

You can earn participation credits in various ways:

- **2 participation points:** additional assignment or check-in besides the letter grade;² initiating threads on the discussion board.
- **1 participation point:** asking questions during class;³ using the discussion board (except initiating threads); using help sessions.
- **0.25 participation point:** per lecture attended.

¹An F is an F.

²Example: Eji only submits 6 WeBWork assignments and 4 problem sets, qualifying for a C. The 4th problem set surplus, since it wasn't used to progress their letter grade, so it gives them 2 engagement credits.

³Procedural questions don't count (e.g. "how do I schedule a meeting with the TA?")

Examples.

- Anna does well on 6 WeBWork assignments and 3 problem sets, checks in a couple times with the TA, and passes her final presentation. She doesn't engage with much with the course at all. She passes with **C-**.
- Ben forgets that he was enrolled in this course, so he speedruns many WeBWork assignments in the end of June, while also checking in a few times with the instructor and the TA, in preparation for the final presentation. He passes with **D**.
- Charlotte can and does attend the online lectures, participates in the discussion boards and has enough free time in the Summer to consume and comment there on additional content related to Multivariable Calculus. They submit the minimum number of satisfactory assignments to obtain an A mark. They pass with **A+**.
- Daniel actually forgets that he was enrolled in the course. He gets an **F**.

ASSIGNMENTS

As outlined in the previous section, the assignments determine your letter grade, and their surplus gives participation points. Note that the final presentation is required to pass the course.

The computer gives grades for WeBWork assignments, and all others will be graded (either by your marker or instructor) as SATISFACTORY or UNSATISFACTORY. The ones that are SATISFACTORY progress your letter grade.

You can submit any form of assignment as many times as you want. However, excluding WeBWork assignments, you can only submit two assignments (problem sets or projects) per week. Each week ends on Sunday, 11:59 pm (Halifax time).

WeBWork assignments. Online homework, accessible through Brightspace from day one. Graded by the computer.

Problem sets. The main source will be exercise bundles posted weekly, but you can also choose-your-own-problems from the textbook, or solve questions from previous years exams. Submissions are on Brightspace. Those with mainly positive feedback will be SATISFACTORY.

Projects. Options will be posted on Brightspace throughout the Spring, many from day one. Submissions are also on Brightspace, according to the their respective project guideline. The submissions conducting the explorations specified in their projects will be SATISFACTORY.

Check-ins. A *check-in* is defined as the act of coming to a help session (see the “Support” section below), solving a problem, and getting feedback on your correctness and clarity. A check-in is always SATISFACTORY.

Final presentation. In the final week of the term (June 22–26) you will book a meeting with your instructor on Teams. In it, you will solve one or two random problems from any of the exercise bundles. The presentation is SATISFACTORY if so are the solutions.

Contact your instructor in case you don’t have access to a writing device such as a tablet.

If you make three UNSATISFACTORY final presentations, then you will need to arrange an in-person proctored exam near your living condition at the end of June. (NOTE: this may incur a fee.) You would pass the exam by scoring at least 50% marks, with your modifiers still determining your final grade.

SUPPORT

Email. You can always reach out to your instructor at daniel.teixeira@dal.ca. Ask anything!

Help sessions. Your instructor and TA can meet with you on Teams to help you with your coursework. Sessions are drop-in or by appointment. The link is the link for the lectures: <https://tinyurl.com/math2001teams>.

- Drop-in sessions: after class until 6 pm (i.e. MWF 5-6 pm)
- Appointments: <https://tinyurl.com/math2001booking>

Learning Centre. Other graduate students are available to help you. Sign up at <https://dal.brightspace.com/d21/home/125201>; hours are M–F 12-4 pm (online).

Discussion Board. There is a forum on Brightspace where you can ask questions and share interesting thoughts. The TA will be contributing to the discussion!

COURSE POLICIES RELATED TO ACADEMIC INTEGRITY

You are encouraged to work with your colleagues, including in the assignments. [Studies have shown](#) that social ties are a main predictor of success in STEM classes. Although you are encouraged to work together, the work you submit must be written by you individually.

There are many online aids to homework at all levels, including AI. If you choose to use such resources, please be cautious: they often provide too detailed an answer, and you may trick yourself into thinking that you understand more than they do by copying those answers. If not for learning, what are we here for?

You can use AI to help with your problems, but remember that your objective is to understand and how to use Multivariable Calculus is, and you may trick yourself into thinking that you understand more than you do by copying answers.

You can also use AI to help with the projects, but you must disclose in detail your use of AI, and you must understand what you are submitting. For instance, you shall use an LLM to help you with the commands and packages to use in a Python project, but you should 1. disclose how did you use the LLM in your submissions, 2. be able to explain all lines in your code to a peer. If a project is suspected to be overly reliant on AI, the student might be asked for clarification or re-submission.

UNIVERSITY POLICIES AND STATEMENTS

Recognition of Mi'kmaq Territory. Dalhousie University would like to acknowledge that the University is on Traditional Mi'kmaq Territory. The Elders in Residence program provides students with access to First Nations elders for guidance, counsel, and support. Visit or e-mail the Indigenous Student Centre at 1321 Edward St or elders@dal.ca. Additional information regarding the Indigenous Student Centre can be found at: https://www.dal.ca/campus_life/communities/indigenous.html

Internationalization. At Dalhousie, 'thinking and acting globally' enhances the quality and impact of education, supporting learning that is "interdisciplinary, cross-cultural, global in reach, and orientated toward solving problems that extend across national borders." Additional internationalization information can be found at: <https://www.dal.ca/about-dal/internationalization.html>

Academic Integrity. At Dalhousie University, we are guided in all our work by the values of academic integrity: honesty, trust, fairness, responsibility, and respect. As a student, you are required to demonstrate these values in all the work you do. The University provides policies and procedures that every member of the university community is required to follow to ensure academic integrity. Additional academic integrity information can be found at: https://www.dal.ca/dept/university_secretariat/academic-integrity.html

Accessibility. The Student Accessibility Centre is Dalhousie's centre of expertise for matters related to student accessibility and accommodation. If there are aspects of the design, instruction, and/or experiences within this course (online or in-person) that result in barriers to your inclusion, please contact the Student Accessibility Centre (https://www.dal.ca/campus_life/academic-support/accessibility.html) for all courses offered by Dalhousie with the exception of Truro. For courses offered by the Faculty of Agriculture, please contact the Student Success Centre in Truro (<https://www.dal.ca/about-dal/agricultural-campus/student-success-centre.html>).

Conduct in the Classroom - Culture of Respect. Substantial and constructive dialogue on challenging issues is an important part of academic inquiry and exchange. It requires willingness to listen and tolerance of opposing points of view. Consideration of individual differences and alternative viewpoints is required of all class members, towards each other, towards instructors, and towards guest speakers. While expressions of differing perspectives are welcome and encouraged, the words and language used should remain within acceptable bounds of civility and respect.

Diversity and Inclusion - Culture of Respect. Every person at Dalhousie has a right to be respected and safe. We believe inclusiveness is fundamental to education. We stand for equality. Dalhousie is strengthened in our diversity. We are a respectful and inclusive community. We are committed to being a place where everyone feels welcome and supported, which is why our Strategic Direction prioritizes fostering a culture of diversity and inclusiveness (Strategic Priority 5.2). Additional diversity and inclusion information can be found at: <http://www.dal.ca/cultureofrespect.html>

Student Code of Conduct. Everyone at Dalhousie is expected to treat others with dignity and respect. The Code of Student Conduct allows Dalhousie to take disciplinary action if students don't follow this community expectation. When appropriate, violations of the code can be resolved in a reasonable and informal manner - perhaps through a restorative justice process. If an informal resolution can't be reached, or would be inappropriate, procedures exist for formal dispute resolution. The full Code of Student Conduct can be found at: https://www.dal.ca/dept/university_secretariat/policies/student-life/code-of-student-conduct.html

Fair Dealing Policy. The Dalhousie University Fair Dealing Policy provides guidance for the limited use of copyright protected material without the risk of infringement and without having to seek the permission of copyright owners. It is intended to provide a balance between the rights of creators and the rights of users at Dalhousie. Additional information regarding the Fair Dealing Policy can be found at: https://www.dal.ca/dept/university_secretariat/policies/academic/fair-dealing-policy.html

Originality Checking Software. The course instructor may use Dalhousie's approved originality checking software and Google to check the originality of any work submitted for credit, in accordance with the Student Submission of Assignments and Use of Originality Checking Software Policy. Students are free, without penalty of grade, to choose an alternative method of attesting to the authenticity of their work and must inform the instructor no later than the last day to add/drop classes of their intent to choose an alternate method. Additional information regarding Originality Checking Software can be found at: https://www.dal.ca/dept/university_secretariat/policies/academic/student-submission-of-assignments-and-use-of-originality-checking-software-policy.html

Student Use of Course Materials. Course materials are designed for use as part of this course at Dalhousie University and are the property of the instructor unless otherwise stated. Third party copyrighted materials (such as books, journal articles, music, videos, etc.) have either been licensed for use in this course or fall under an exception or limitation in Canadian Copyright law. Copying this course material for distribution (e.g. uploading to a commercial third-party website) may lead to a violation of Copyright law.

STUDENT RESOURCES AND SUPPORT

University Policies and Programs.

- Important Dates in the Academic Year (including add/drop dates): http://www.dal.ca/academics/important_dates.html
- Classroom Recording Protocol: https://www.dal.ca/dept/university_secretariat/policies/academic/classroom-recording-protocol.html
- Dalhousie Grading Practices Policies: https://www.dal.ca/dept/university_secretariat/policies/academic/grading-practices-policy.html
- Grade Appeal Process: https://www.dal.ca/campus_life/academic-support/grades-and-student-records/appealing-a-grade.html
- Sexualized Violence Policy: https://www.dal.ca/dept/university_secretariat/policies/health-and-safety/sexualized-violence-policy.html
- Scent-Free Program: <https://www.dal.ca/dept/safety/programs-services/occupational-safety/scent-free.html>

Learning and Support Resources.

- General Academic Support - Advising (Halifax): https://www.dal.ca/campus_life/academic-support/advising.html
- General Academic Support - Advising (Truro): <https://www.dal.ca/about-dal/agricultural-campus/ssc/academic-support/advising.html>
- Student Health & Wellness Centre: https://www.dal.ca/campus_life/health-and-wellness.html
- On Track (help you transition into university, and supports you through your first year at Dalhousie and beyond): https://www.dal.ca/campus_life/academic-support/On-track.html
- Indigenous Student Centre: https://www.dal.ca/campus_life/communities/indigenous.html
- Indigenous Connection: <https://www.dal.ca/about-dal/indigenous-connection.html>
- Elders-in-Residence (The Elders in Residence program provides students with access to First Nations elders for guidance, counsel, and support. Visit the office in the Indigenous Student Centre or contact the program at elders@dal.ca or 902-494-6803): <https://cdn.dal.ca/content/dam/dalhousie/pdf/academics/UG/indigenous-studies/Elder-Protocol-July2018.pdf>
- Black Student Advising Centre: https://www.dal.ca/campus_life/communities/black-student-advising.html
- International Centre: https://www.dal.ca/campus_life/international-centre.html
- South House Sexual and Gender Resource Centre: <https://southhousehalifax.ca/about/>
- LGBTQ2SIA+ Collaborative: <https://www.dal.ca/dept/vpei/edia/education/community-specific-spaces/LGBTQ2SIA-collaborative.html>
- Dalhousie Libraries: <http://libraries.dal.ca/>
- Copyright Office: <https://libraries.dal.ca/services/copyright-office.html>
- Dalhousie Student Advocacy Services: <https://www.dsu.ca/dsas?rq=student%20advocacy>
- Dalhousie Ombudsperson: https://www.dal.ca/campus_life/safety-respect/student-rights-and-responsibilities/where-to-get-help/ombudsperson.html
- Human Rights and Equity Services: <https://www.dal.ca/dept/hres.html>
- Writing Centre: https://www.dal.ca/campus_life/academic-support/writing-and-study-skills.html
- Study Skills/Tutoring: http://www.dal.ca/campus_life/academic-support/study-skills-and-tutoring.html
- Faculty of Science Advising Support: <https://www.dal.ca/faculty/science/current-students/undergrad-students/degree-planning.html>

Safety.

- Biosafety: <http://www.dal.ca/dept/safety/programs-services/biosafety.html>
- Chemical Safety: <https://www.dal.ca/dept/safety/programs-services/chemical-safety.html>
- Radiation Safety: <http://www.dal.ca/dept/safety/programs-services/radiation-safety.html>
- Laser Safety: <https://www.dal.ca/dept/safety/programs-services/radiation-safety/laser-safety.html>